

Calibration and correction of spatial distortions in 2D detector systems

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Many photon and other radiation 2D detection systems introduce spatial distortions into recorded data. As long as these distortions are to some degree repeatable it is possible to use a grid of holes of known positions to illuminate the detector and calibrate the distortion. Knowledge of this distortion can be used to correct subsequent images. Aspects of calibration grid design are discussed with the aim of achieving optimum results. Cross-correlation with known hole functions is proposed as a method for estimating grid peak centres giving optimum accuracy and efficiency. Strategies to search efficiently complete grids are also presented. Where a physical model of the distortion process is lacking or inadequate, bi-cubic splines are proposed as suitable interpolating functions for defining approximate spatial distortion functions. Bi-cubic splines allow both adequate flexibility to describe arbitrary distortions and, most importantly, to allow efficient calculation for the correction of data. Calibration and correction algorithms have been developed and applied to 2D detector systems. Sample results are presented.

1. Introduction

Real detector systems may introduce distortions to measured data: intensity non-linearity, signal independent noise, positional spatial distortions, non-uniformity in intensity response from one pixel to another, and blurring owing to convolution with a point spread function which may be a function of detector position. (We refer to a detection system, as distortions may well be introduced at various stages in the detection and recording processes.)

The distortions may be reduced provided that they are sufficiently constant and predictable between a calibration test and the acquisition of scientific data. Care must be taken as the conditions of a calibration test must be the same or sufficiently similar to the conditions under which the desired data are taken e.g. Distortions may be energy, temperature, or distance dependent, depending on the type and design of the detection system. Here we are primarily concerned with the positional spatial distortions which may be highly regular owing to well understood physical design of the detector e.g. field distortions in a TV camera or image intensifier, or may be much more irregular owing to poorly understood mechanical irregularities e.g. movement of a scanning bed of an image plate scanner.

This work was initiated when significant distortions were found in "Imaging Plate" detector system data [1], and has been designed to be flexible enough to be generally applicable to spatial distortions from (cartesian) 2D detector systems. In particular a new detector consisting of an X-ray image intensifier detector (XRII) optically coupled to a high resolution

CCD camera is under development at the ESRF [2,3]. This detector system can suffer from large spatial distortions owing to its convex surface and electric field geometry.

Many examples of spatial distortion calibration and correction exist (e.g. [4–6]), however optimum methods for obtaining high positional accuracy or for improving computational efficiency, have been little discussed. Efficiency may or may not be of concern when calibrating spatial distortions but will generally be important when applying corrections to subsequent data. Efficiency is considered important for calibration in the general case, as users will be discouraged from making new calibration measurements, even when necessary, if the procedure takes too long. Efficiency will become of even greater importance for larger area detectors which are presently being proposed. Here we aim to achieve an arbitrarily high positional accuracy in the most efficient manner.

2. Calibration grid design

Typically an opaque calibration mask of a regular 2D grid of transparent holes is used to produce a known pattern on the detector. The mask is placed as close as is possible to the detector and illuminated with radiation. Using small holes ("pin-holes") allows the distortion between an ideal 2D position and its measured position to be determined.

The design of the calibration grid clearly has an important influence on the quality of the calibration and corrections which may be made. Different aspects of the grid itself are discussed:

2.1. Choice of calibration pattern

It is unlikely that a grid of holes is always optimum for calibrating a particular type of detector system and/or data. Nevertheless, for the general case we will assume that some pattern of pin-holes is appropriate.

The time spent calibrating the distortion will be approximately proportional to the number of holes. Thus it is interesting to consider the pattern of holes which allows sufficiently accurate correction of the spatial distortion with the minimum number of holes.

Regular patterns of pin-holes are generally used. For a particular detector system distortion it is likely that some irregular pattern would be optimum (for the same number of holes) i.e. the holes would be more numerous at those points where the distortion is expected to be greatest or least understood. In the case of a detector system where spatial distortions in the X and Y direction are completely orthogonal, it would only be necessary to use two perpendicular lines of holes instead of a 2D grid. The hole pattern could also be made optimum as a function of the data e.g. where the data are of no importance, there is no requirement to correct their positions, and hence no need to have holes.

In the general case of unknown data and unknown detector system distortion characteristics, a regular grid is the best choice. This corresponds to existing practises. However, existing calibration grids tend to be 2D cartesian grids. In the general case hexagonal grids would be superior as their sampling of spatial frequencies is better, or in other words, the maximum distance from a calibrated position is minimised for a given number of holes.

2.2. Mask grid spacing

The optimum grid spacing will depend on the physical understanding of the distortion, the characteristic distance of the distortion, and the accuracy required. If the distortions are essentially random then the grid must be fine enough to allow adequate sampling of the distortion function. The Nyquist criterion suggests that the sampling frequency should be twice the highest frequency in the distortion function (defined as measured position minus ideal position as a function of measured position). If we could determine the distortion function this method could theoretically be used to design an ideal calibration mask and apply ideal corrections. In practice the distortion function may contain very high spatial frequencies which cannot be sampled by a practical mask, leading to some inaccuracy in the correction function. Under these conditions the necessary grid spacing will be determined by the acceptable inaccuracy in the corrected

images.

In calculating the centre of each hole (in section 3.1.1) it will be required for efficiency reasons that the holes recorded by the detector system be well separated from each other. This places a minimum limit on the grid spacing which will depend on the detector point spread function and on the size of each hole.

The “optimum” mask is a function of both the detector system characteristics and the required data, however, with the aim of producing a general system it is possible to produce a finely spaced grid, but only use some regular sub-set of the holes for distortion calibration and removal.

2.3. Hole size and hole shape

For accurate determination of hole centres circular holes are the best shape (see section 3.1.1). The difficulties in accurately manufacturing other hole shapes will also favour circular holes.

The holes must be small enough to allow the spots to be recorded well separated on the detector. Thomas [6] shows that the theoretical uncertainty in finding a peak position in a particular direction δx , e.g. X or Y -direction from an ideal hole of radius a with N photons is:

$$\delta x = a/2\sqrt{N}. \quad (1)$$

For a calibration measurement with a fixed flux per unit area at the calibration mask, N is proportional to the hole area which is proportional to a^2 for an ideal detector. Eq. (1) suggests that the accuracy is independent of the hole area. However, under real conditions the mask will not be totally opaque, the detector will suffer from noise, and the hole will be convoluted with the detector system Point Spread Function (PSF) [7].

Here we consider the effect of the detector system PSF on the accuracy with which the peak centre may be determined. The spot recorded by the detector $h(x, y, a, \sigma)$, defined in cartesian co-ordinates x, y , with grid hole radial size a and detector PSF standard deviation σ , is the convolution of the ideal hole $f(x, y, a)$ with the detector PSF $g(x, y, \sigma)$, which we assume is Gaussian:

$$f(x, y, a) = \begin{cases} 1 & \text{if } 0 \leq \sqrt{x^2 + y^2} \leq a, \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

$$g(x, y, \sigma) = \frac{1}{2\pi\sigma^2} \exp \left[-\frac{x^2 + y^2}{2\sigma^2} \right], \quad (3)$$

$$\begin{aligned} h(x', y', a, \sigma) \\ = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y, a) g(x - x', y - y', \sigma) dx dy \end{aligned}$$

$$= \frac{1}{2\pi\sigma^2} \times \int_{-a}^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \exp \left[-\frac{(x-x')^2 + (y-y')^2}{2\sigma^2} \right] dy dx. \quad (4)$$

Since $f(x, y, a)$ and $g(x, y, \sigma)$ are both radially symmetric, $h(x, y, a, \sigma)$ is also radially symmetric and may be defined in polar co-ordinates r, θ , where:

$$x' = r \cos \theta; \quad y' = r \sin \theta. \quad (5)$$

$h(r, \theta, a, \sigma)$ is independent of θ , and we choose for convenience $\theta = 0$.

$$\begin{aligned} h(r, a, \sigma) &= \frac{1}{2\pi\sigma^2} \\ &\times \int_{-a}^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \exp \left[-\frac{(r-x)^2 + y^2}{2\sigma^2} \right] dy dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-a}^a \frac{\text{erf} \left[\frac{\sqrt{a^2-x^2}}{\sqrt{2}\sigma} \right]}{\exp \left[\frac{(r-x)^2}{2\sigma^2} \right]} dx. \end{aligned} \quad (6)$$

Following the arguments of Thomas, the theoretical uncertainty in finding the peak position in any radial direction, $\delta r(a, \sigma, N)$ can be shown to be:

$$\begin{aligned} \delta r(a, \sigma, N)^2 &= \frac{\langle r^2 \rangle}{N} \\ &= \frac{1}{N} \frac{\int_0^{+\infty} r^2 h(r, a, \sigma) 2\pi r dr}{\int_0^{+\infty} h(r, a, \sigma) 2\pi r dr}. \end{aligned} \quad (7)$$

Eq. (7) is solved numerically to produce values for the positional uncertainty for given a, σ, N . As $\sigma \rightarrow 0$ Eq. (7) gives the same results as Eq. (1).

N is still proportional to a^2 , but now the positional uncertainty is a complicated function of a and σ . The meaning of Eq. (7) can be understood by considering the effect of different hole sizes relative to the standard deviation of the PSF, comparing the accuracy relative with a system with no PSF. These results are independent of N . Fig. 1 shows the relative uncertainty in position compared to a system with a delta function detector PSF, against hole size relative to detector PSF.

We conclude that the grid hole diameters should be $> 6\sigma$ of the detector system PSF.

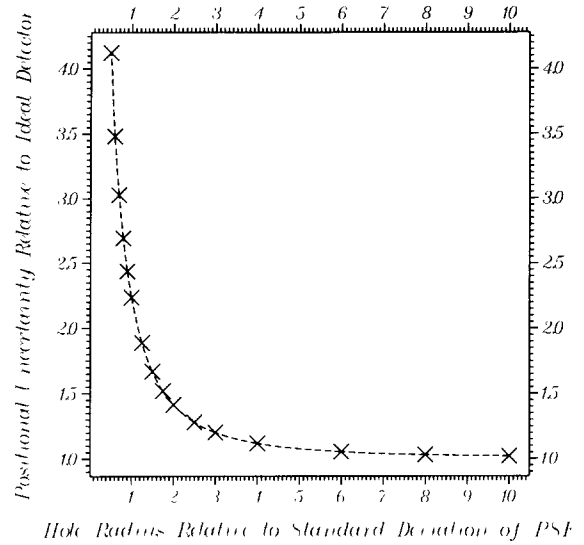


Fig. 1. Relative positional uncertainty compared to a detector system with no PSF plotted for grid holes of sizes from 0.5 to 10.0 times the detector PSF standard deviation

3. Spatial distortion calibration and correction

The calibration and correction process can be subdivided into the following operations:

- (i) Peak search: find the centres of the grid peaks.
- (ii) Distortion definition: calculate values of the distortion at each measured grid peak.
- (iii) Distortion fitting with interpolating function: find an interpolating function which fits closely (or exactly) to the distortion as measured at the grid peaks.
- (iv) Distortion correction: estimation of distortion at all pixel positions, and re-binning or application of an inverse method.

Each operation may be performed using a variety of techniques with the most appropriate technique varying for different detector systems e.g. the type of interpolating function.

3.1. Peak search

As a calibration grid may contain thousands of grid holes, the method used to perform the peak search should be reasonably efficient. The peak search may be further sub-divided into the method used to find the best centre of each peak and the method used to search the image for further peaks.

3.1.1. Peak centre determination

Typically peak centres may be determined to sub-pixel accuracy using iterative profile fitting techniques. Whilst potentially giving optimum results this is very computationally demanding. Given the

following conditions, identical accuracy may be obtained using cross-correlation techniques which generally require much less computation:

- The peaks are separated.
- The peak profiles are symmetric.
- Individual peaks on plane backgrounds (may be inclined in 2D).
- (Known peak shape).

For a known peak profile, cross-correlation of the measured data with the known profile is optimum for finding the peak centre [9]. (The centre is at the position where the cross-correlation value is a maximum). The constraint that the peak shape is known is very weak as, so long as a reasonable guess may be used as a starting point, it is possible to “learn” the real peak shape by summing the functions found after the initial centres have been found. The new peak template can be used to re-calculate the peak centres, and re-calculate the profile template. This operation is continued iteratively until convergence.

If the peak profile is identical for all peaks then the calculation is much simpler. Otherwise, it will be necessary to take account of the change in peak profile. If this is not well understood some loss of accuracy must occur and a bias may appear in the predicted positions.

The method works as follows (the rationale is explained below):

- (i) The known peak profile is calculated over a suitable sized template with the same resolution (approximate) as the data.
- (ii) The mean of the calculation template is subtracted from every pixel in the template.
- (iii) The template is cross-correlated with the data around the region of interest.
- (iv) The position at which the cross-correlation is a maximum is the centre of the peak.

If the peaks and hence the template function are centro-symmetric, then the sine components of its Fourier transform will all be zero. If the background consists of a linear slope it contains only dc and sine components. Convolution with the template function will filter out the sloping background completely if the dc component is also zero. Thus by choosing a symmetric peak template and subtracting the mean value from the template prior to cross-correlation, we can find optimum peak centres even on sloping backgrounds [8]. In the 2D situation the peak must be circularly symmetric for the filter to be perfect. (More complicated backgrounds can still lead to inaccuracies in the estimated peak centres, but under normal experimental conditions such inaccuracies will be extremely small and other effects will probably lead to greater inaccuracies.)

If the profile is wrong, but both the true profile and the assumed profile are circularly symmetric then, on

average, the centre will be correct but the variance in position will be greater than for the ideal profile.

To obtain accuracy to greater than one pixel it is necessary to calculate a template at a finer resolution and to perform over-sampling on the data i.e. each pixel in the data is duplicated a number of times in each direction in a larger array. The cross-correlation is performed as before. This method allows the trade-off between position accuracy and computational time to be easily controlled.

3.1.2. 2D grid search strategies

If the distortion function is sampled sufficiently to fulfil or nearly to fulfil the Nyquist criterion, it will be smooth and the position of each peak may be estimated with reasonable accuracy from the positions of previously measured peaks. By estimating the positions of all the peaks from previous ones, starting from an initial three, much computation may be avoided by only calculating the cross-correlation at a few positions.

If the estimated positions are reasonably accurate, computation may be reduced by using an “ascent” method to find the optimum centre. For each starting position the cross-correlation values are calculated for a 3×3 square. If the maximum cross-correlation is in the centre of the square the optimum centre has been found, otherwise the search is moved in the direction of the maximum cross-correlation value, and the next 3×3 square is considered.

If the estimated positions are not very accurate it will be more efficient to find the centre in successive stages. First a peak centre is found to the nearest pixel, then each pixel is sub-divided into four sub-pixels and the peak centre is found to the nearest half a pixel. After the nearest pixel centre is known, it is only necessary to calculate the cross-correlation over four by four data pixels to find the best sub-pixel centre. The sub-division process is continued until the desired accuracy is obtained.

These two strategies can be combined in an algorithm which alters its search technique based on the success of previous centre predictions.

3.2. Distortion definition

The definition of distortion values is complicated by rotation of the calibration grid with respect to the detector raster. Even with careful alignment, rotation can lead to differences between ideal position position and rotated position of similar size to the distortion. This rotation should not be counted as “distortion” so we need a method estimate the rotation and hence calculate the distortion separately. Unfortunately this rotation will also be distorted. Estimation of the rotation angle is further complicated by non-square pixels

as, in general, the two dimensions of the average pixel may be different.

The average pixel dimensions can be calculated from the number of pixels measured between grid holes spaced widely apart, to minimise the effect of local distortions. Alternatively the RMS value may be taken from all adjacent holes. Provided the grid is reasonably aligned to the detector, the distance between holes in rows can be used for the X -direction and the distance between holes in columns can be used for the Y -direction.

The best strategy for deducing the rotation will be detector system specific e.g. If a detector system suffers minimum X -distortion along a vertical line in the middle of the detector, and minimum Y -distortion along a horizontal line in the middle of the vertical, then using lines of holes near to these lines should be optimal for deducing the distortion. In general, the strategy of looking at distant points should be close to optimal. Averaging over all distant points allows a rotation angle to be calculated which should cancel out long range distortions.

From the calculated rotation angle, the known grid hole spacings and the required ideal pixel dimensions, the values of the distortion function for the X and Y -directions may be calculated based on an arbitrary zero distortion position.

3.3. Distortion fitting with interpolating functions

It is necessary to use an interpolating function to estimate the distortion for all detector positions not only where the spots have been measured. The function may be based on a physical model of the detection process or may be empirical using some interpolation method (e.g. linear, spline). Any function based on a physical model of the system must be detector system dependent so little can be said generally. For such a method to work the physics of the detection process must be understood to the required accuracy of corrections.

If no distortion model is available or is poorly understood some prior knowledge can be used to interpolate positions away from those of spots. If nothing is known of the distortion function, linear interpolation between known distortion points is the most “non-committal” approach. If the function is “known” to be smooth between spot points then bi-cubic spline interpolation may be more appropriate. For real data a number of different functions can be used in parallel on the same data-sets to see which produces the best results in subsequent data processing.

For unknown smooth distortions we propose using bi-cubic spline interpolation, but recognise that where the distortion has long range coherence, polynomials may be more appropriate. The choice of the bi-cubic

splines is mainly motivated by efficiency considerations for correcting distorted data, however, splines may be seen to have some potentially useful properties;

- Arbitrary shaped local distortions may be fitted and the distortions in different areas of the detector may be unrelated.

- Computational efficiency, as the calculation is locally defined. (Using higher order polynomials would require much more calculation.)

- Defining spline knots on regular or irregular 2D grids allows the relevant knot points for any position to be efficiently located. (Using interpolation to the actual measured positions would require very time consuming calculation of appropriate triangulation points.)

Two spline surfaces need to be defined: one for the distortion in the X -direction as a 2D function of measured position, and one for the distortion in the Y -direction.

It may seem inaccurate to fit the spline function only approximately to the measured distortions, but the estimated positions have errors and in practice some positions may be erroneous owing to grit partially blocking the hole. If the errors in estimated distortion values are independent and normally distributed, then fitting a spline surface to the measured distortion values with a RMS discrepancy equal to the average positional error is optimum. The spline surface can be kept constrained by the data by choosing splines with far fewer knot points than measured peak positions, and confidence may be given to the interpolated values between peak positions.

3.4. Distortion correction

Ideally the “knowledge” of the introduced distortions would be incorporated as part of further data analysis (this is done in the powder diffraction data analysis program PLATYPUS [10]), or even as part of a complete inverse method where some physical system is modelled constrained by the measured (and distorted) data. For practical reasons it is often desirable to recreate an “undistorted” 2D image. Much existing data analysis software (human brain) assumes as input a regular flat 2D detector.

Once the spatial distortion interpolating function has been defined, the correction process is relatively simple. The interpolating function allows the value of the distortion for all points on the distorted images to be estimated. Thus the position of distorted pixels on a corrected pixel image may be calculated. The intensity in each input pixel may be re-distributed to the corrected pixels in proportion to covered area. Whilst the input pixels may be translated to arbitrary shapes on the corrected image, they may be approximated

by rectangles. The problem of re-binning pixel data from a distorted grid to a regular grid is covered by Stanton [5]. He discusses approximate techniques for re-binning and trade-offs between accuracy and efficiency.

The approach taken by Stanton is the most reasonable a priori approach in the general situation. It should be noted that this re-binning stage causes a blurring to the image. We could reduce this blurring by putting all the intensity from one pixel into this corrected centre position at the expense of unevenness in the corrected image.

4. Other corrections

Spatial distortion is only one of a number of distortions which may be introduced by a detector system. In general these other distortions will need to be corrected and it is important to realise that they are inter-dependent.

In particular any intensity non-linearity correction will need to be performed before any re-binning is performed to correct spatial distortion. Re-binning an image from a distorted grid to a corrected one will change the effective areas of pixels and so lead to a change in the uniformity of response (flat field response). This effect can be corrected in software at the time of re-binning since it is purely dependent on the geometrical areas of the distorted pixels which are added to each corrected pixel. Alternatively it is also possible to perform an uniformity calibration measurement and re-bin this measurement to the corrected pixel grid prior to using it for flat field correction.

5. Practical calibration measurements

The ideas presented above have been implemented as part of the program FIT2D [11]. Calibration masks have been specified and manufactured, and calibration procedures developed for different detectors.

To measure the spatial distortion, two nickel grid masks (35×43 cm) were mainly used, a fine grid with 2 mm spaced $\varnothing 200 \mu\text{m}$ holes, and a coarse grid with 4 mm spaced $\varnothing 600 \mu\text{m}$ holes^{#1}. The grids were made by electro-forming. The accuracy of the grids was verified by optical measurements (grid step errors $< 20 \mu\text{m}$, orthogonality errors $< 0.04^\circ$, although the accuracy is probably much better). Both these grids are cartesian, which is not optimum in the general case, but they were designed to calibrate a detector system

where the distortions in the X and Y -directions are almost completely orthogonal. Additionally some copper grid masks were used for the XR11 system to avoid magnetic effects from nickel.

The scanned images were processed using the program FIT2D. Peak positions were found and X and Y -distortions were estimated using the techniques described above.

6. Results of calibration measurements

The algorithms presented have now been applied to three different types of detector system and to five different designs. Here we present the results necessary to determine the appropriateness of this approach for a particular detector system. These results should only be taken as examples as the distortions present in different detectors of the same model may be different.

The distortion is categorised in two types: pixel non-squareness, and local distortion. Pixel non-squareness may be a problem for some analysis software which assumes square pixels, but in many cases it is accounted for and does not need to be corrected.

6.1. Commercial imaging plate scanner

Two different commercial imaging plate scanners were investigated for distortion and stability. Here we report results from one of these systems. An X-ray tube with a tungsten target running at 30 kV 10 mA was used for exposing the grid pattern onto the detector. The detector was placed 3 m from the tube where the beam measured 20×20 cm. The grid was held tightly against the detector by a set of magnets which, together with the large distance from the source, minimised parallax errors. The detector was exposed sufficiently to give good statistics in the peaks whilst having very low intensity in the background.

For the imaging plate scanner system investigated, the average pixel non-squareness was 1.1% for a fine resolution mode, and 0.23% for a coarser resolution mode. The average pixel dimensions were found to be very repeatable from image to image. These figures compare with about 0.2% typically for scanned X-ray film data.

Since the local distortion should not include the pixel non-squareness, the ideal pixel dimensions were set to the estimated average pixel dimensions. The reference point (zero distortion) is arbitrary, and FIT2D allows the user to define the peak position where the distortion is assumed to be zero. We have used the centre of the detector as the zero point.

Figs. 2 and 3 show examples of the distortion as measured by lines of holes in the X and Y directions.

^{#1} Supplied by Attewell Ltd, 4 Southbridge Way, Southall, Middlesex UB2 4BY, UK.

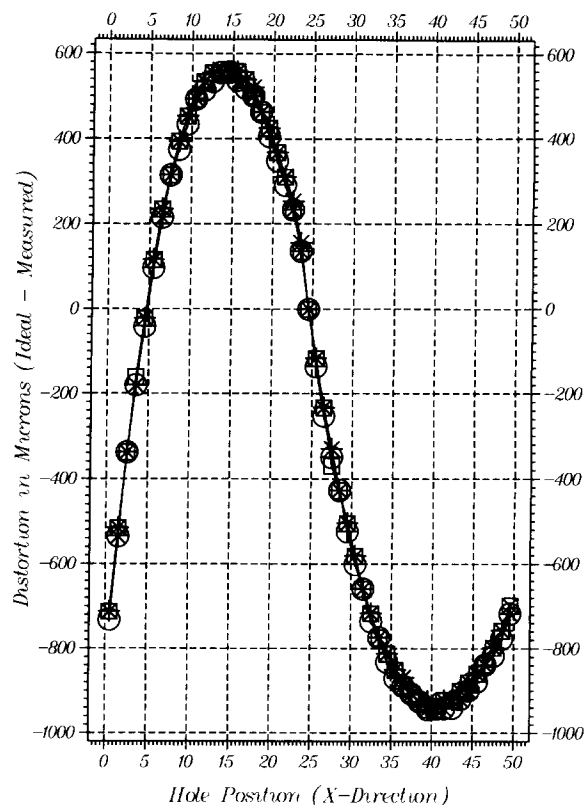


Fig. 2. *X*-distortion versus *X*-position plotted for 5 consecutive grid rows.

The different lines on each graph show the distortion as measured by consecutive lines of grid holes. It should be noted that a symmetric change in pixel widths over the detector surface leads to an anti-symmetric distortion. (The software allows the whole 2D measured surface distortion to be displayed as a false colour image.) The variation in the measured distortions from one curve to another gives a measure of the accuracy of the positions, although there is also systematic change of the *X*-distortion with change in *Y*-position and systematic change of the *Y*-distortion with change in *X*-position.

Self consistency tests were performed which showed that the grid images themselves could be corrected with a RMS error of less than 10 μm starting from a RMS error of 540 μm . However, this result is of limited significance owing to non-repeatability in the detector distortion from one image to the next. Trying to perform highly accurate corrections is not valid if the distortion itself has changed between the calibration image and scientific data images. To determine the likely success of corrections and the frequency at which new calibration measurements would need to be performed, a long series of distortion

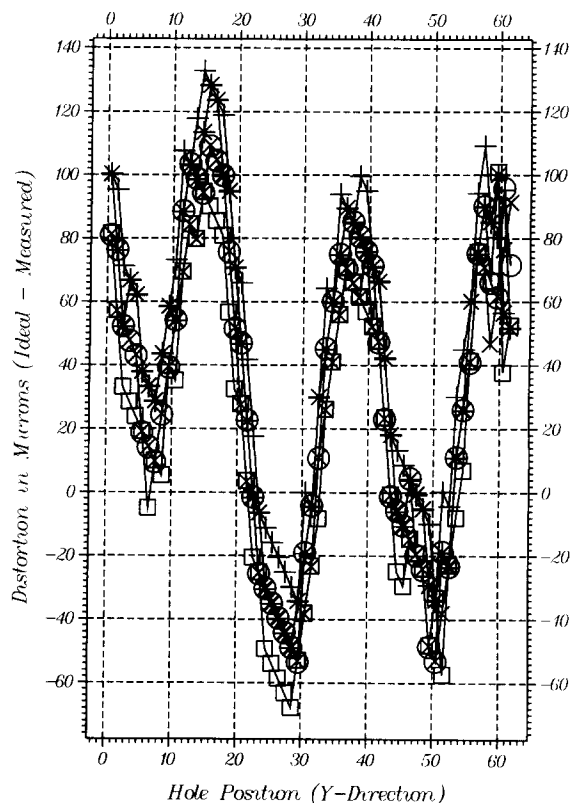


Fig. 3. *Y*-distortion versus *Y*-position plotted for 5 consecutive grid columns.

measurements were performed to investigate the non-repeatability.

If the distortion were perfectly repeatable, then the difference between two distortion measurements would be zero. Differences between distortion measurements show either errors in the measurements or changes in the distortion. By looking at the RMS difference the measurement errors will be minimised

Table 1
Summary of Spatial Distortion Results

Resolution/ Direction	RMS uncorrected distortion [μm]	RMS distortion repeatability [μm]
Fine <i>X</i> -direction	382 ± 7	20 ± 7
Fine <i>Y</i> -direction	136 ± 15	67 ± 44
Coarse <i>X</i> -direction	540 ± 9	30 ± 10
Coarse <i>Y</i> -direction	151 ± 16	75 ± 41

Table 2
Fine Y-direction scan spatial distortion repeatability (RMS μm)

Run	1	2	3	4	5	6	7	8	9	10
Day	0	1	1	2	2	8	8	10	10	31
1	*	15.0	15.0	12.3	19.4	65.3	76.7	132.3	132.3	35.3
2		*	7.9	17.6	14.1	59.1	66.1	125.2	125.2	43.2
3			*	14.1	9.7	54.7	62.6	123.5	123.5	45.0
4				*	14.1	57.3	66.1	126.1	126.1	43.2
5					*	48.5	57.3	117.3	117.3	51.2
6						*	16.8	78.5	79.4	92.6
7							*	70.6	71.4	99.7
8								*	9.7	149.9
9									*	147.3

and a good measure of the non-repeatability will be obtained. This figure is not a direct measure of the remaining position error in the corrected data, but should give a good estimate of it. There are factors, both positive and negative, which mean that the remaining errors in the corrected data will be slightly different from the RMS non-repeatability.

Table 1 shows the uncorrected RMS distortion estimates in microns together with the RMS distortion repeatability estimates. The figures for RMS uncorrected distortion and its standard deviation are taken from a number of repeated calibration measurements. The non-repeatability measurements are the average of the RMS differences between all combinations of different calibration measurements. The error estimates are not strictly standard deviations since the individual results are correlated, however they give a reasonable measure of the variation.

The results show that the distortion can potentially be greatly reduced in the X-direction to a level which is very useful for diffraction experiments. Measurements taken over a month show that there is no significant change in the distortion (compared to differences from measurements taken minutes apart). In the Y-direction the situation is more complicated. The average repeatability is much poorer but nevertheless shows that on average spatial distortion corrections based on previous calibration measurements reduce the spatial distortion. However the variation in results is very large. Table 2 shows the RMS differences between individual measurements and each of the other measurements, for the fine resolution Y-direction distortion. It is evident that there is large random change between measurements i.e. some consecutive measurements are very similar whereas others are very different. We conclude that the system suffers from changes which may occur at any time.

Inspection of the detector system mechanism suggests that a simple mechanical change should be able

to both reduce the distortion in the Y-direction and make it much more repeatable. We are presently in contact with the manufacturers with the aim of applying this modification.

6.2. X-ray image intensifier detector

The XRII suffers from spatial distortion of a rather different nature to the imaging plate system. The spatial distortion is mainly due to the geometry of the detector, but may also be due to distortion in the electro-optics of the image intensifier. The X-ray detection surface is a nearly spherical and curves away from the X-ray source.

A protective entrance window means that a calibration mask cannot be placed directly on the detection surface and so parallax cannot be eliminated. The distortion is thus source distance and position dependent, and calibration measurements must be made with an X-ray source at the same position as the X-ray source for experimental data. In the case of X-ray crystallography this is the sample position. By illuminating a grid placed at the entrance window and calibrating its distorted image, we may correct subsequent data to a flat detector at the same distance from the source as the mask.

As the image intensifier stage is extremely susceptible to magnetic fields, it is important to perform the calibration measurements in the same experimental configuration as the scientific data. Also the calibration mask must be non-magnetic.

We used a $\varnothing$ 310 mm XRII (model TH9432HX) in the full field mode, optically coupled to a 1024×1024 CCD camera. From calibration measurements taken with the source at 27 cm from the entrance window RMS distortion values were $1850 \mu\text{m}$ for the X-direction and $1490 \mu\text{m}$ for the Y-direction. Whilst these values are very large the distortion is very smooth and this largely corresponds to the short

source to detector distance.

Again it is the non-repeatability of distortion which can limit the effectiveness of distortion correction. A number of measurements show this to be less than 50 μm (third of a pixel) in the X or Y directions over eighteen hours, but these measurements suffered from poor statistics and a poor grid. Separate calibration measurements with a more accurate grid and much better statistics show the RMS non-repeatability to be less than 8 μm (twentieth of a pixel) over 6 h.

Work is continuing to understand the effect of changes in the electro-magnetic environment on the image distortion.

7. Application to physics data analysis

The calibration measurement with the grid pattern only reveals the spatial distortion at some discrete number of points, i.e. where the holes in the grid are located. The distortion is not known between two successive holes in the grid pattern, and the overall success of the correction procedure can only be estimated from the non-repeatability measurements. The best test of the method is to analyse real data and see how well the data fit to the theory.

X-ray crystallography diffraction data have been collected at both the ESRF and Daresbury synchrotrons using imaging plate detector systems. The samples analysed were tetragonal lysozyme, magnesium formate, and concanavalin [12]. Prior to correction, unsuccessful attempts were made to analyse a number of data images. The positions of recorded diffraction peaks on the scan were too different to the expected positions as predicted by the analysis programs. Corrected 2D images were produced which could be used as input to standard crystallographic analysis programs.

The analysis programs produce many indicators as to the performance of the analysis and the quality of the data. A summary of the positional errors measured as reduced χ^2 statistics or as RMS errors is given in

Table 3
Comparison of reduced χ^2 and RMS values for uncorrected and corrected data

		Uncorrected	Corrected
Lysozyme	$\chi^2 x$	27.9	2.3
	$\chi^2 y$	4.9	1.6
Magnesium	$\chi^2 x$	16.2	1.0
	$\chi^2 y$	4.7	3.3
Concavalin A	RMS	95 μm	45 μm

Table 3 for uncorrected and corrected data^{#2}. The improved fit of the data is clear from the improved χ^2 and from the improved RMS fit.

The more reproducible correction in the X -direction of the data is demonstrated by the large improvement of fit in this direction as compared to the weak improvement in the other direction.

8. Conclusion

The process of calibrating and correcting spatial distortions using calibration masks has been examined in detail. Criteria for designing optimum calibration masks and algorithms for efficiently analysing calibration images have been proposed. The algorithms have been developed with the experience of real calibration measurements and take account of real life problems such as mask rotation and missing mask holes. A program (FIT2D) has been developed by one of us allowing detector systems to be calibrated and corrections applied to images. The program allows the different stages in the calibration and correction processes to be treated separately allowing flexibility for application to different detector systems.

The calibration and analysis procedures have been applied in an extended study of the properties of one detector system and have shown that separate calibration measurements may be used to correct spatial distortion and considerably improve data quality. They have also shown a limitation of the present detector system and led to the project of minor modification of the system to both reduce the distortion and improve its repeatability. With this modification the detector system should be capable of producing high quality scientific results.

Other detector systems have also been examined, showing that the algorithms developed may be used with different types of distortion function.

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^{#2} The χ^2 statistic output by the analysis program results from user input estimates of positional inaccuracy, but also reflects errors owing to intensity non-linearity, saturated pixels, and spot profile errors. It is non-trivial to convert these χ^2 values into RMS distance errors.

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References

- [1] Y. Amemiya, T. Matsushita, A. Nakagawa, Y. Satow, J. Miyahara and J. Chikawa, Nucl. Instr. and Meth. A 266 (1988) 645.
- [2] J.P. Moy and J. Morse, 3rd London Conf. on Position Sensitive Detectors, (1993)
- [3] J.P. Moy and S. Gibney, Proc. SPIE 1982, Photoelectric Detection and Imaging, 1993 (1993) pp. 17–23.
- [4] P.E. Kinahan and J.S. Karp, Nucl. Instr. and Meth. A 299 (1990) 484.
- [5] M. Stanton, W.C. Philips, Y. Li and K. Kalata, J. Appl. Cryst. 25 (1992) 549.
- [6] D.J. Thomas, Proc. Roy. Soc. Lond. A 425 (1989) 129.
- [7] D. Bourgeois, J.P. Moy, O. Svensson and Å. Kvik, submitted to J. Appl. Cryst.
- [8] M. Wahl, M.Sc. Thesis, School of Information Science and Technology, Liverpool Polytechnic (1991).
- [9] D.J. Thomas, J. Phys. (Paris) Suppl. 8 Tome 47 (1986).
- [10] R. Piltz, private communication.
- [11] A.P. Hammersley, FIT2D Reference Manual, ESRF Internal Report, EXP/AH/92-02 (1993).
- [12] S.O. Svensson, A.P. Hammersley, A. Thompson, A. Gonzalez, and T. Ursby, CCP4/EPS-EACBM Newsletter 29 (1993).